

OS PRACTICAL

PRACTICAL2

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution. Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

OUTPUT:

```
Rashmika@LAPTOP-UNQ0CN  x  +  v  -  □  ×
Rashmika@LAPTOP-UNQ0CNLA:~$ cd OS_LAB
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB$ cd Practical
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical$ cd Practical2
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical2$ nano Practical2.c
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical2$ gcc Practical2.c -o
Practical2
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical2$ nano sample.txt
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical2$ ./Practical2 sample.
txt copy.txt
File copied successfully.
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical2$ cat copy.txt
Hey whatsapp
this is Practical2
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical2$ strace --version
strace -- version 6.19
Copyright (c) 1991-2026 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx
32-mpers
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical2$ strace cat sample.tx
t
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffd648770c8 /* 26 vars */)
= 0
brk(NULL) = 0x56c05f346000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x74c2d5ba2000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directo
ry)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27891, ...}) = 0
mmap(NULL, 27891, PROT_READ, MAP_PRIVATE, 3, 0) = 0x74c2d5b9b000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"...,
832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x74c2d5b68
000
```

```
mmap(0x74c2d5b6f000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_D  
DENYWRITE, 3, 0x7000) = 0x74c2d5b6f000  
mmap(0x74c2d5b90000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,  
3, 0x28000) = 0x74c2d5b90000  
mmap(0x74c2d5b97000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D  
ENYWRITE, 3, 0x2f000) = 0x74c2d5b97000  
mmap(0x74c2d5b99000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_A  
NONYMOUS, -1, 0) = 0x74c2d5b99000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEX  
EC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...,  
832) = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0  
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x74c2d5b3a  
000  
mmap(0x74c2d5b3e000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_  
DENYWRITE, 3, 0x4000) = 0x74c2d5b3e000  
mmap(0x74c2d5b62000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,  
3, 0x28000) = 0x74c2d5b62000  
mmap(0x74c2d5b66000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D  
ENYWRITE, 3, 0x2c000) = 0x74c2d5b66000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC)  
= 3  
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...,  
832) = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0  
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x74c2d5a1  
4000  
mmap(0x74c2d5a26000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_  
DENYWRITE, 3, 0x12000) = 0x74c2d5a26000  
mmap(0x74c2d5ac6000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,  
3, 0xb2000) = 0x74c2d5ac6000  
mmap(0x74c2d5b38000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D  
ENYWRITE, 3, 0x123000) = 0x74c2d5b38000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC)  
= 3  
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0"...,  
832) = 832
```

```
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..  
., 840, 64) = 840  
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0  
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..  
., 840, 64) = 840  
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x74c2d560  
0000  
mmap(0x74c2d5628000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_  
_DENYWRITE, 3, 0x28000) = 0x74c2d5628000  
mmap(0x74c2d57c0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,  
3, 0x1c0000) = 0x74c2d57c0000  
mmap(0x74c2d580e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_  
_DENYWRITE, 3, 0x20d000) = 0x74c2d580e000  
mmap(0x74c2d5814000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_  
_ANONYMOUS, -1, 0) = 0x74c2d5814000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcrc2-8.so.0", O_RDONLY|O_CLO  
EXEC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...,  
832) = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0  
mmap(NULL, 701112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x74c2d5968  
000  
mmap(0x74c2d596b000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_  
_DENYWRITE, 3, 0x3000) = 0x74c2d596b000  
mmap(0x74c2d59e7000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,  
3, 0x7f000) = 0x74c2d59e7000  
mmap(0x74c2d5a12000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D  
ENYWRITE, 3, 0xa9000) = 0x74c2d5a12000  
close(3) = 0  
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0  
x74c2d5966000  
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =  
0x74c2d5963000  
arch_prctl(ARCH_SET_FS, 0x74c2d5963b00) = 0  
set_tid_address(0x74c2d5964128) = 959  
set_robust_list(0x74c2d5963de0, 24) = 0  
rseq(0x74c2d59636a0, 0x20, 0, 0x53053053) = 0  
mprotect(0x74c2d580e000, 16384, PROT_READ) = 0  
mprotect(0x74c2d5a12000, 4096, PROT_READ) = 0  
mprotect(0x74c2d5b38000, 4096, PROT_READ) = 0
```

```
mprotect(0x56c053b41000, 1462272, PROT_READ) = 0
mprotect(0x74c2d5be7000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\xf5\xf4\xf6\xf1\xa3\xdb\x79\x83", 8, GRND_NONBLOCK) = 8
munmap(0x74c2d5b9b000, 27891) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0xffa8feb2, 0xc3c3350c]}, f_namelen=255, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NODEV|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7ffd37b8f130) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x56c05f346000
brk(0x56c05f367000) = 0x56c05f367000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbdf"..., 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-"..., 1024) = 1024
read(3, "e 0 0\ndepts /dev/pts devpts rw,"..., 1024) = 1024
read(3, "osuid,nodev,noexec,relatime,nosy"..., 1024) = 1024
read(3, "s /mnt/wslg/run/user/1001 tmpfs "..., 1024) = 118
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Time out)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x74c2d5645cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
```

```
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "56c0531d8000-56c0532ca000 r--p 0"... , 1024) = 1024
read(3, "0 08:30 13644" "..." , 1024) = 1024
read(3, "00 08:30 13647" "..." , 1024) = 1024
read(3, " /usr/lib/x86_64-lin"..." , 1024) = 1024
read(3, "0 13641" /us"..." , 1024) = 566
close(3) = 0
sched_getaffinity(959, 32, [0 1 2 3 4 5 6 7 8 9 10 11]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x74c2d5b9f000
mprotect(0x74c2d5b9f000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x74c2d5ba0000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 959
rt_sigaction(SIGSEGV, {sa_handler=0x56c0534fd0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x74c2d5645cb0}, NULL, 8) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=0x56c0534fd0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x74c2d5645cb0}, NULL, 8) = 0
prctl(PR_GET_AUXV, 0x7ffd37b8ec20, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...}) = 0
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFD
```

```

Rashmika@LAPTOP-UNQ0CN  ×  +  ▾
TATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFD
IR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CL
OEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTO
RER|SA_RESTART, sa_restorer=0x74c2d5645cb0}, {sa_handler=SIG_IGN, sa_mask=[P
IPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x74c2d5645cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=S
A_RESTORER, sa_restorer=0x74c2d5645cb0}, {sa_handler=0x56c0534fd0c0, sa_mask
=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x74c2d5645cb0}
, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA
_RESTORER, sa_restorer=0x74c2d5645cb0}, {sa_handler=0x56c0534fd0c0, sa_mask=
[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x74c2d5645cb0},
8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOS
T|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXT
EN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x56c05f388000) = 0x56c05f388000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=ST
ATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=31,
...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=31, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 31
splice(4, NULL, 1, NULL, 31, 0Hey whatsapp
this is Practical2
) = 31
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x74c2d5b9f000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
Rashmika@LAPTOP-UNQ0CNLA:~/OS_LAB/Practical/Practical2$ |

```